Francesco Paolo Maiale

Academic curriculum vitae
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Research profile

Applied mathematician working in numerical linear algebra, numerical PDEs and variational analysis. Research connects structured spectral algorithms, fractional diffusion and dynamical systems with spectral shape optimization and free-boundary regularity.

Academic appointment

September 2023–present | Postdoctoral researcher in Mathematics, Gran Sasso Science Institute, Italy. Research in analysis, calculus of variations, numerical PDEs and numerical linear algebra. Current appointment runs to 31 August 2027.

Education

November 2018–November 2022 | PhD in Mathematics, Scuola Normale Superiore, Pisa. Awarded 7 November 2022, cum laude. Thesis: Shape Optimization Problems and Regularity of the Free Boundaries. Supervisor: Giuseppe Buttazzo.

August 2016–June 2018 | Master's degree in Mathematics, University of Pisa. Awarded 8 June 2018, 110/110 cum laude. Thesis: Linking Theorems, Kato-Ponce Inequalities, and Applications in Mathematical Physics. Supervisor: Vladimir Georgiev.

August 2013–September 2016 | Bachelor's degree in Mathematics, University of Pisa. Awarded 16 September 2016, 110/110 cum laude. Thesis: Wiener's Lemma.

Teaching experience

November 2023–present (teaching annually, November–February) | Co-lecturer, first-year doctoral Numerical Analysis, GSSI. MATLAB-based computational teaching and selected theoretical lectures, with Nicola Guglielmi and colleagues. Responsible for the practical MATLAB examination, worth half the total grade; typical cohort 8–12 students.

2018–2020 | Tutor, Complementi di Matematica, Scuola Normale Superiore. Exercise sessions for groups of two to four first-year students, approximately two hours per week, extending topics from Analysis I.

September 2017–September 2018 | Teaching Assistant, Analysis I, University of Pisa. Mechanical engineering students, with Giuseppe Buttazzo. Exercise sessions, office hours, written and oral examinations, and examination-committee membership.

Research funding and affiliation

Research supported through MUR PRIN 2022 Advanced numerical methods for time dependent parametric partial differential equations with applications and PRIN-PNRR FIN4GEO. Role: postdoctoral researcher supported by collaborative projects. Affiliated with INdAM–GNCS, the National Group for Scientific Computing.

Published and accepted research

1. F. P. Maiale. A note on harmonic polynomials on Heisenberg and Carnot groups. Journal of Mathematical Analysis and Applications, accepted (2026). arXiv:2602.02200.

2. N. Guglielmi and F. P. Maiale. A Gripenberg-Like Algorithm for the Lower Spectral Radius. SIAM Journal on Matrix Analysis and Applications 47(1) (2026), 77–106. doi:10.1137/24M1672456.

3. G. Buttazzo, F. P. Maiale, D. Mazzoleni, G. Tortone and B. Velichkov. Regularity of the optimal sets for a class of integral shape functionals. Archive for Rational Mechanics and Analysis 248, 44 (2024). doi:10.1007/s00205-024-01984-y.

4. F. P. Maiale, G. Tortone and B. Velichkov. The boundary Harnack principle on optimal domains. Annali della Scuola Normale Superiore di Pisa, Classe di Scienze 25(1) (2024), 127–149. doi:10.2422/2036-2145.202112_003.

5. K. O. Idu and F. P. Maiale. $C^{1,\alpha}$-rectifiability in low codimension in Heisenberg groups. Analysis and Geometry in Metric Spaces 12(1) (2024), article 20230105. doi:10.1515/agms-2023-0105.

6. F. P. Maiale, G. Tortone and B. Velichkov. Epsilon-regularity for the solutions of a free boundary system. Revista Matemática Iberoamericana 39(5) (2023), 1947–1972. arXiv:2108.03606.

7. G. Buttazzo, F. P. Maiale and B. Velichkov. Shape optimization problems in control form. Rendiconti Lincei. Matematica e Applicazioni 32(3) (2021), 413–435. doi:10.4171/RLM/942.

8. K. O. Idu, V. Magnani and F. P. Maiale. Characterizations of k-rectifiability in homogeneous groups. Journal of Mathematical Analysis and Applications 500(2) (2021), article 125120. doi:10.1016/j.jmaa.2021.125120.

9. G. Buttazzo and F. P. Maiale. Shape optimization problems for functionals with a boundary integral. Journal of Convex Analysis 28(2) (2021), 429–456. arXiv:2007.11317.

10. G. Buttazzo and F. P. Maiale. Optimal one-dimensional structures for the principal eigenvalue of two-dimensional domains. Nonlinear Analysis 191 (2020). arXiv:2007.05725.

11. V. Georgiev and F. P. Maiale. Critical points of Dirac functional with broken symmetry. Pliska Studia Mathematica 29 (2018), 11–20.

Public preprints

12. F. P. Maiale. Perturbation from symmetry for semiconvex lower-order partial-wave nonlinear Dirac equations (2026). Submitted. arXiv:2607.14679.

13. F. P. Maiale, A. Trofimova and A. De Marinis. Approximation bounds for norm constrained deep neural networks (2025). Submitted. arXiv:2512.20422.

14. F. P. Maiale, A. Trofimova and N. Guglielmi. Probabilistic Analysis of the Random Spectral Radius for a Matrix Family (2025). Submitted. arXiv:2511.16271.

Academic service

11–13 February 2026 | Local organizer, Advanced Numerical Methods for Parametric PDEs, GSSI.

26–28 January 2026 | Co-organizer, Mathematics of data driven models, GSSI.

Journal refereeing: Journal of Mathematical Analysis and Applications; Automatica.

Selected talks

October 2026 | Fast fractional diffusion in 2D/3D. NIC60, international conference in honor of Nicola Guglielmi.

July 2026 | Fast fractional diffusion in 2D/3D. CMAM 2026, Vienna.

June 2024 | Existence and regularity of the minima of integral shape functionals. Barcelona Analysis Conference.

May 2024 | An algorithm for computing the lower spectral radius. 50 Anni di Calcolo delle Variazioni, Pisa.

November 2023 | Existence and regularity of the minima of integral shape functionals. ASAAP Seminar, University of L'Aquila/GSSI.

Additional research training and honors

August each year, 2017–2020 | Jyväskylä International Summer School. Courses in computational verification, shape optimization and free boundaries, Lie groups, geometric control, regularity, stochastic homogenization, optimal transport and stochastic calculus of variations.

July–August 2012 | Gran Sasso–Princeton Physics Summer School. Ranked first in the national selection competition. Two-week program at Princeton University.

Educational resources

Mathematical lecture notes publicly available at francescomaiale.github.io/teaching/, covering topics from undergraduate, master's and doctoral study.

Languages and computational skills

Italian: native. English: C1, Cambridge English: Advanced, 2018.

Scientific computing: MATLAB, Python and Fortran. Scientific writing: advanced LaTeX. Code profile: github.com/francescomaiale.